

Question:

Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Reasoning + Answer:

Let's break down the problem step by step to determine how much profit Josh made from flipping the house .

1. Initial Cost of the House :

Josh buys the house for \$ 80 , 000 .

$$\text{Initial cost} = \$80,000$$

2. Cost of Repairs :

Josh spends an additional \$ 50 , 000 on repairs .

$$\text{Cost of repairs} = \$50,000$$

3. Total Investment :

To find the total amount Josh invested in the house , we add the initial cost and the cost of repairs .

$$\text{Total investment} = \text{Initial cost} + \text{Cost of repairs}$$

Substituting the values :

$$\text{Total investment} = \$80,000 + \$50,000$$

$$\text{Total investment} = \$130,000$$

4. Increase in Value :

The problem states that the value of the house increased by 150% after the repairs . To find the new value of the house , we need to calculate 150% of the total investment and add it to the total investment .

5. Calculating 150% of the Total Investment :

First , we convert 150% to a decimal :

$$150\% = 1.5$$

Next , we multiply the total investment by 1.5 to find the increase in value :

$$\text{Increase in value} = 1.5 \times \$130,000$$

Performing the multiplication :

$$1.5 \times 130,000 = 1.5 \times (130 \times 1,000)$$

$$1.5 \times 130 = 195$$

$$195 \times 1,000 = 195,000$$

6. New Value of the House :

To find the new value of the house , we add the increase in value to the total investment :

$$\text{New value} = \text{Total investment} + \text{Increase in value}$$

Substituting the values :

$$\text{New value} = \$130,000 + \$195,000$$

$$\text{New value} = \$325,000$$

7. Calculating the Profit :

The profit is the difference between the new value of the house and the total investment :

$$\text{Profit} = \text{New value} - \text{Total investment}$$

Substituting the values :

$$\text{Profit} = \$325,000 - \$130,000$$

$$\text{Profit} = \$195,000$$

Thus , the profit Josh made from flipping the house is :

<think>

1. Initial cost = \$ 80 , 000

2. Cost of repairs = \$ 50 , 000

3. Total investment = \$ 80 , 000 + \$ 50 , 000 = \$ 130 , 000

4. Increase in value = 1.5 * \$ 130 , 000 = \$ 195 , 000

5